Lecture 13

Finite-dimensional existence and classification

Objectives. *For each $\lambda \in P_+$, an explicit quotient of $M(\lambda)$ is finite-dimensional and simple. This completes the classification begun in Lecture 12.*

By the end of this lecture you will be able to:

- Write the quotient presentation of $L(\lambda)$ for $\lambda \in P_+$.
- Prove local finiteness and Weyl invariance for the simple-root $\mathfrak{sl}_2$ actions.
- Prove that the quotient is finite-dimensional and isomorphic to $L(\lambda)$.
- Prove the bijection between P_+ and the isomorphism classes of finite-dimensional simple modules.

Throughout, $\mathfrak{g}$ is a complex semisimple Lie algebra with Cartan subalgebra $\mathfrak{h}$, simple roots $\alpha_1, \dots, \alpha_\ell$ and Weyl group W, as in Lecture 9. For $\mathfrak{g} \neq 0$, Verma modules are infinite-dimensional.

13.1 The quotient presentation

For each i fix $e_i \in \mathfrak{g}_{\alpha_i}$ and $f_i \in \mathfrak{g}_{-\alpha_i}$ with $h_i = [e_i, f_i] = \alpha_i^\vee$. Then

$$[h_i, e_i] = 2e_i, \qquad [h_i, f_i] = -2f_i, \qquad [e_i, f_i] = h_i.$$

The span $\mathfrak{s}_i$ of e_i, h_i and f_i is a copy of $\mathfrak{sl}_2$ (Lecture 8). Write

$$Q_+ = \sum_{i=1}^{\ell} \mathbb{Z}_{\geq 0}\alpha_i, \qquad \mu \leq \lambda \iff \lambda - \mu \in Q_+.$$

A weight μ lies in P exactly when every $\mu(h_i)$ is an integer. The simple reflections act by

$$s_i\mu = \mu - \mu(h_i)\alpha_i.$$

We use the following facts from earlier lectures.

(F1) For $i \neq j$, $\alpha_j - \alpha_i$ is not a root, so $[e_j, f_i] = 0$. The e_i generate $\mathfrak{n}_+$. To see this, write a nonsimple positive root as $\beta = \gamma + \delta$ with $\gamma, \delta \in R_+$ (Lecture 9). Then $[\mathfrak{g}_\gamma, \mathfrak{g}_\delta] = \mathfrak{g}_\beta$ by Lecture 8. Induction on height shows that each $\mathfrak{g}_\beta$ with $\beta \in R_+$ lies in the subalgebra generated by the e_i.

(F2) $M(\lambda) = U(\mathfrak{n}_-)v_\lambda$ has weights in $\lambda - Q_+$, finite-dimensional weight spaces and $M(\lambda)_\lambda = \mathbb{C}v_\lambda$. PBW monomials applied to v_λ form a basis. If a module V is generated by $u \in V_\lambda$ with $\mathfrak{n}_+ u = 0$, then $v_\lambda \mapsto u$ extends to a surjection $M(\lambda) \to V$. The module $M(\lambda)$ has a unique maximal proper submodule, so every simple quotient is isomorphic to $L(\lambda)$ (Lecture 12).

(F3) W is finite and generated by the s_i. Every $\mu \in P$ is W-conjugate to an element of P_+. For $\lambda \in P_+$, the set $\{\mu \in P_+ : \mu \le \lambda\}$ is finite (Lecture 9).

(F4) Finite-dimensional modules of $\mathfrak{g}$, and of $\mathfrak{sl}_2$, are completely reducible (Lecture 5). The finite-dimensional simple $\mathfrak{sl}_2$-modules are the V_n of Lecture 6.

Lemma 13.1 (An $\mathfrak{sl}_2$ identity · *P*)**.** *In $U(\mathfrak{s}_i)$, for $k \ge 1$,*

$$e_i f_i^k = f_i^k e_i + k\, f_i^{k-1}(h_i - k + 1).$$

Proof. The relation $h_i f_i = f_i(h_i - 2)$ and induction on r give

$$h_i f_i^r = f_i^r (h_i - 2r) \qquad (r \ge 0).$$

Write the commutator as a telescoping sum. Then use $[e_i, f_i] = h_i$, the relation above with $r = k - 1 - j$, and $\sum_{j=0}^{k-1} 2(k-1-j) = k(k-1)$:

$$\begin{aligned} e_i f_i^k - f_i^k e_i &= \sum_{j=0}^{k-1} f_i^{\,j}\,[e_i, f_i]\, f_i^{\,k-1-j} \\ &= \sum_{j=0}^{k-1} f_i^{\,j} h_i f_i^{\,k-1-j} \\ &= \sum_{j=0}^{k-1} f_i^{\,k-1}\big(h_i - 2(k-1-j)\big) \\ &= k\, f_i^{k-1} h_i - k(k-1) f_i^{k-1} \\ &= k\, f_i^{k-1}(h_i - k + 1). \end{aligned}$$

■

Definition 13.2 (The candidate module)**.** Let $\lambda \in P_+$ and $m_i = \lambda(h_i) \in \mathbb{Z}_{\ge 0}$. Put

$$N(\lambda) = \sum_{i=1}^{\ell} U(\mathfrak{g})\, f_i^{m_i+1} v_\lambda, \qquad V(\lambda) = M(\lambda)/N(\lambda).$$

Let u be the image of v_λ in $V(\lambda)$.

A *singular vector* is a nonzero weight vector w with $\mathfrak{n}_+ w = 0$.

Proposition 13.3 (Singular vectors and a nonzero quotient · *P*)**.** *Each $w_i = f_i^{m_i+1} v_\lambda$ satisfies $\mathfrak{n}_+ w_i = 0$. Moreover $N(\lambda)_\lambda = 0$, so $u \ne 0$ and $V(\lambda)_\lambda = \mathbb{C}u$.*

Proof. By PBW, w_i is a nonzero weight vector:

$$w_i \in M(\lambda)_{\lambda - (m_i+1)\alpha_i}.$$

For $j \ne i$, (F1) gives $e_j f_i = f_i e_j$, so

$$e_j w_i = f_i^{m_i+1} e_j v_\lambda = 0.$$

For $j = i$, Lemma 13.1 with $k = m_i + 1$ gives

$$\begin{aligned} e_i w_i &= f_i^{m_i+1} e_i v_\lambda + (m_i+1) f_i^{m_i} (h_i - m_i) v_\lambda \\ &= 0 + (m_i+1)(m_i - m_i) f_i^{m_i} v_\lambda = 0. \end{aligned}$$

The annihilator of w_i in $\mathfrak{g}$ is a subalgebra. It contains every e_j, so it contains $\mathfrak{n}_+$ by (F1).

PBW gives the triangular factorisation (Lecture 11)

$$U(\mathfrak{g}) = U(\mathfrak{n}_-)U(\mathfrak{h})U(\mathfrak{n}_+).$$

Here $\mathfrak{h}$ and $\mathfrak{n}_+$ preserve $\mathbb{C}w_i$. Hence

$$U(\mathfrak{g})w_i = U(\mathfrak{n}_-)U(\mathfrak{h})U(\mathfrak{n}_+)w_i = U(\mathfrak{n}_-)w_i.$$

This space is spanned by weight vectors whose weights μ satisfy

$$\lambda - \mu \in (m_i+1)\alpha_i + Q_+.$$

The coefficient of α_i in $\lambda - \mu$ is at least 1, so $\mu \neq \lambda$.

The λ-weight space of a sum of weight submodules is the sum of their λ-weight spaces. Thus

$$N(\lambda)_\lambda = \sum_{i=1}^{\ell} \big(U(\mathfrak{g})w_i\big)_\lambda = 0.$$

Hence $u \neq 0$, and $V(\lambda)_\lambda$ is the image $\mathbb{C}u$ of $M(\lambda)_\lambda = \mathbb{C}v_\lambda$. ∎

Since $N(\lambda)$ is spanned by weight vectors, $V(\lambda)$ is the direct sum of its weight spaces. These weight spaces are finite-dimensional by (F2). Since $\lambda \in P$, the weights lie in $\lambda - Q_+ \subseteq P$. The module $V(\lambda)$ is generated by u, and

$$\mathfrak{n}_+ u = 0, \qquad f_i^{m_i+1} u = 0 \quad (1 \le i \le \ell).$$

13.2 The simple-root $\mathfrak{sl}_2$ actions

Lemma 13.4 (Moving a power past $x \cdot P$). *For $x, y \in \mathfrak{g}$ and $N \ge 0$, in $U(\mathfrak{g})$,*

$$y^N x = \sum_{j=0}^{N} \binom{N}{j} \big((\operatorname{ad} y)^j x\big)\, y^{N-j}.$$

Proof. Let L_y and R_y be left and right multiplication by y on $U(\mathfrak{g})$. Associativity and the definition of $\operatorname{ad} y$ give

$$L_y R_y = R_y L_y, \qquad \operatorname{ad} y = L_y - R_y.$$

Hence $\operatorname{ad} y$ commutes with R_y, and the binomial theorem applies:

$$L_y^N = (R_y + \operatorname{ad} y)^N = \sum_{j=0}^{N} \binom{N}{j} R_y^{N-j} (\operatorname{ad} y)^j.$$

Apply both sides to x. ∎

Proposition 13.5 (Local nilpotence · *P*). *For each i and each $w \in V(\lambda)$, there is k with $e_i^k w = 0$ and $f_i^k w = 0$.*

Proof. Let $y = e_i$ or $y = f_i$, and put $\mathfrak{g}_0 = \mathfrak{h}$. Then $\operatorname{ad} y$ maps $\mathfrak{g}_\beta$ into $\mathfrak{g}_{\beta+\alpha_i}$ or $\mathfrak{g}_{\beta-\alpha_i}$, respectively. Only finitely many β have $\mathfrak{g}_\beta \neq 0$, so there is r with

$$(\operatorname{ad} y)^r = 0 \quad \text{on } \mathfrak{g}.$$

Since $y^k w = 0$ implies $y^{k'} w = 0$ for $k' \geq k$, the set

$$V' = \{w \in V(\lambda) : y^k w = 0 \text{ for some } k \geq 0\}$$

is a subspace. Suppose $y^k w = 0$ and $x \in \mathfrak{g}$. Apply Lemma 13.4 with $N = k + r$. Each term has $j \geq r$, so $(\operatorname{ad} y)^j x = 0$, or $k + r - j \geq k$, so $y^{k+r-j} w = 0$. Hence

$$y^{k+r} x w = \sum_{j=0}^{k+r} \binom{k+r}{j} ((\operatorname{ad} y)^j x) y^{k+r-j} w = 0.$$

So $xw \in V'$, and V' is a submodule.

The submodule V' contains u. For $y = e_i$ this holds since $e_i u = 0$, and for $y = f_i$ since $f_i^{m_i+1} u = 0$. As u generates $V(\lambda)$, we get $V' = V(\lambda)$. Take the larger of these two exponents. ∎

Proposition 13.6 (Local finiteness · *P*). *Every $w \in V(\lambda)$ lies in a finite-dimensional $\mathfrak{s}_i$-submodule spanned by weight vectors.*

Proof. Every w is a finite sum of weight vectors, and a finite sum of such submodules is again one. So we may assume that w has weight μ. By Proposition 13.5, choose A and B with

$$e_i^{B+1} w = 0, \qquad f_i^{A+1} e_i^b w = 0 \quad (0 \leq b \leq B).$$

Let

$$X = \operatorname{span}\{f_i^a e_i^b w : \ 0 \leq a \leq A, \ 0 \leq b \leq B\}.$$

Each spanning vector is a weight vector:

$$f_i^a e_i^b w \in V(\lambda)_{\mu + (b-a)\alpha_i}.$$

So X is $\mathfrak{h}$-stable. It is f_i-stable by the choice of A.

For stability under e_i, put $c = (\mu + b\alpha_i)(h_i)$, so $h_i e_i^b w = c\, e_i^b w$. For $a \geq 1$, Lemma 13.1 gives

$$\begin{aligned} e_i f_i^a e_i^b w &= f_i^a e_i^{b+1} w + a\, f_i^{a-1} (h_i - a + 1) e_i^b w \\ &= f_i^a e_i^{b+1} w + a(c - a + 1) f_i^{a-1} e_i^b w. \end{aligned}$$

Both terms lie in X, and the first is zero when $b = B$. For $a = 0$ the vector is $e_i^{b+1} w$, again in X or zero. Thus X is an $\mathfrak{s}_i$-submodule. ∎

Proposition 13.7 (Weyl invariance · *P*). *For every weight μ and every $\sigma \in W$,*

$$\dim V(\lambda)_{\sigma\mu} = \dim V(\lambda)_\mu.$$

Proof. Each s_i maps P to P, since $\alpha_i \in P$. So W permutes P. Every weight of $V(\lambda)$ lies in $\lambda - Q_+ \subseteq P$. If $\mu \notin P$, then $\sigma\mu \notin P$, and both weight spaces vanish. Let $\mu \in P$.

The s_i generate W by (F3), so it suffices to treat $\sigma = s_i$. Write $V = V(\lambda)$. Then

$$k = \mu(h_i) \in \mathbb{Z}, \qquad s_i\mu = \mu - k\alpha_i.$$

Replacing μ by $s_i\mu$ if necessary, assume $k \geq 0$. We show that

$$f_i^k : V_\mu \to V_{\mu - k\alpha_i}, \qquad e_i^k : V_{\mu - k\alpha_i} \to V_\mu$$

are injective.

Let $x \neq 0$ be a weight vector of weight μ or $\mu - k\alpha_i$. By Proposition 13.6, x lies in a finite-dimensional $\mathfrak{s}_i$-submodule X. By (F4),

$$X = X_1 \oplus \cdots \oplus X_s, \qquad X_t \cong V_{n_t}.$$

Write $x = x_1 + \cdots + x_s$ with $x_t \in X_t$. The projections onto the X_t commute with h_i. So each nonzero component x_t of x is an h_i-eigenvector with eigenvalue k or $-k$, respectively.

In V_n use the basis $v_j = f_i^j v_0$ of Lecture 6. It satisfies

$$\begin{aligned} h_i v_j &= (n - 2j)v_j, \\ f_i v_j &= v_{j+1} \quad (j < n), \\ e_i v_j &= j(n - j + 1)v_{j-1} \quad (j > 0). \end{aligned}$$

A k-eigenvector in V_n is a multiple of v_j with $n - 2j = k$. A $(-k)$-eigenvector is a multiple of $v_{j'}$ with $n - 2j' = -k$. In this case put $j = j' - k = n - j' \geq 0$. Then again $n - 2j = k$, and the $(-k)$-eigenvector is a multiple of v_{j+k}. In both cases $j + k = n - j \leq n$, so

$$\begin{aligned} f_i^k v_j &= v_{j+k} \neq 0, \\ e_i^k v_{j+k} &= \prod_{t=j+1}^{j+k} t(n - t + 1)\, v_j \neq 0, \end{aligned}$$

since $1 \leq t \leq n$ in the product. The vectors $f_i^k x_t$, or $e_i^k x_t$, are nonzero and lie in distinct summands. Hence $f_i^k x \neq 0$ in the first case and $e_i^k x \neq 0$ in the second. The two injections give

$$\dim V_\mu \leq \dim V_{s_i\mu} \leq \dim V_\mu.$$

■

13.3 Finite dimension and irreducibility

> **Lemma 13.8** (Cyclic modules · *P*). *Let V be a finite-dimensional $\mathfrak{g}$-module that is the direct sum of its weight spaces. Suppose $\dim V_\lambda = 1$ and V is generated by a vector $u \in V_\lambda$. Then V is simple.*

Proof. By (F4), $V = V^{(1)} \oplus \cdots \oplus V^{(s)}$ with simple submodules $V^{(t)}$. The projections onto the summands commute with $\mathfrak{h}$, so

$$V_\lambda = V_\lambda^{(1)} \oplus \cdots \oplus V_\lambda^{(s)}.$$

Since $\dim V_\lambda = 1$, exactly one summand, say $V^{(1)}$, has $V_\lambda^{(1)} \neq 0$. Then

$$u \in V_\lambda = V_\lambda^{(1)}, \qquad V = U(\mathfrak{g})u \subseteq V^{(1)}.$$

So $V = V^{(1)}$ is simple. ■

Theorem 13.9 (Existence · *P*). *Let $\lambda \in P_+$. The module $V(\lambda)$ is finite-dimensional, and $V(\lambda) \cong L(\lambda)$. In particular, $L(\lambda)$ is finite-dimensional, and $N(\lambda)$ is the maximal proper submodule of $M(\lambda)$.*

Proof. Finitely many weights. Let μ be a weight of $V(\lambda)$. Then $\mu \in P$, and (F3) gives $\sigma \in W$ with $\sigma\mu \in P_+$. By Proposition 13.7, $\sigma\mu$ is also a weight, so $\sigma\mu \le \lambda$. Write $\operatorname{wt} V(\lambda)$ for the set of weights. Then

$$\operatorname{wt} V(\lambda) \subseteq \bigcup_{\nu \in P_+,\ \nu \le \lambda} W\nu.$$

The index set is finite by (F3), and W is finite. So there are finitely many weights.

Finite dimension. Each weight space of $V(\lambda)$ is a quotient of a weight space of $M(\lambda)$. By (F2),

$$\dim V(\lambda) = \sum_{\mu \in \operatorname{wt} V(\lambda)} \dim V(\lambda)_\mu \le \sum_{\mu \in \operatorname{wt} V(\lambda)} \dim M(\lambda)_\mu < \infty.$$

Irreducibility. By Proposition 13.3, Lemma 13.8 applies to $V(\lambda)$ and u. So $V(\lambda)$ is a simple quotient of $M(\lambda)$, and (F2) gives $V(\lambda) \cong L(\lambda)$. Its kernel $N(\lambda)$ is therefore the unique maximal proper submodule. ∎

13.4 The classification

Theorem 13.10 (Classification · *P*). *The assignment $\lambda \mapsto L(\lambda)$ induces a bijection from P_+ to the set of isomorphism classes of finite-dimensional simple $\mathfrak{g}$-modules.*

Proof. Well defined. This is Theorem 13.9.

Surjective. By Lecture 12, every finite-dimensional simple module is isomorphic to $L(\lambda)$ for some $\lambda \in P_+$.

Injective. An isomorphism commutes with $\mathfrak{h}$, so it preserves weights. The weights of $L(\lambda)$ lie in $\lambda - Q_+$, and λ is one of them. If $L(\lambda) \cong L(\mu)$, then $\mu \in \lambda - Q_+$ and $\lambda \in \mu - Q_+$. Since the α_i are linearly independent,

$$\lambda - \mu \in Q_+ \cap (-Q_+) = \{0\}.$$ ∎

Simple ideals

Decompose $\mathfrak{g}$ into simple ideals $\mathfrak{g}_k$ (Lecture 4). Choose a Cartan subalgebra $\mathfrak{h}_k \subseteq \mathfrak{g}_k$ and positive roots for each k. Write

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_r, \qquad \mathfrak{h} = \mathfrak{h}_1 \oplus \cdots \oplus \mathfrak{h}_r.$$

This $\mathfrak{h}$ is toral, since $x = \sum_k x_k$ with $x_k \in \mathfrak{h}_k$ acts diagonalisably on $\mathfrak{g}_k$ by $\operatorname{ad} x_k$. It is also maximal. Indeed, a toral, hence abelian, subalgebra containing $\mathfrak{h}$ lies in

$$C_{\mathfrak{g}}(\mathfrak{h}) = \bigoplus_k C_{\mathfrak{g}_k}(\mathfrak{h}_k) = \mathfrak{h},$$

by the self-centraliser theorem [C] of Lecture 7 (Humphreys, §8.2), applied to each maximal toral $\mathfrak{h}_k$ of the semisimple $\mathfrak{g}_k$. So $\mathfrak{h}$ is a Cartan subalgebra of $\mathfrak{g}$, and we use it from now on.

For $j \ne k$, $\mathfrak{h}_j$ acts by zero on $\mathfrak{g}_k$. Hence the roots of $\mathfrak{g}$ are the roots of the $\mathfrak{g}_k$, extended by zero. Choose their union as the positive system of $\mathfrak{g}$. The simple roots and coroots of $\mathfrak{g}$ are the disjoint unions of those of the $\mathfrak{g}_k$, so

$$\mathfrak{h}^* = \mathfrak{h}_1^* \oplus \cdots \oplus \mathfrak{h}_r^*, \qquad P_+ = P_+^{(1)} \times \cdots \times P_+^{(r)}.$$

For $\mathfrak{g}_k$-modules (V_k, ρ_k), the *outer tensor product* $V_1 \boxtimes \cdots \boxtimes V_r$ is $V_1 \otimes \cdots \otimes V_r$ with $x = x_1 + \cdots + x_r$, with $x_k \in \mathfrak{g}_k$, acting by

$$\sum_{k=1}^{r} \mathrm{id} \otimes \cdots \otimes \rho_k(x_k) \otimes \cdots \otimes \mathrm{id}\,.$$

The terms for different k commute, and $[\mathfrak{g}_j, \mathfrak{g}_k] = 0$ for $j \neq k$. So this is a representation.

Theorem 13.11 (Factorisation · *P*). *For $\lambda = (\lambda^1, \dots, \lambda^r) \in P_+$,*

$$L(\lambda) \cong L(\lambda^1) \boxtimes \cdots \boxtimes L(\lambda^r), \qquad \dim L(\lambda) = \prod_{k=1}^{r} \dim L(\lambda^k).$$

Every finite-dimensional simple $\mathfrak{g}$-module has this form.

Proof. Let T be the outer tensor product, and let v_k be a highest weight vector of $L(\lambda^k)$. A tensor of weight vectors of weights μ^k has weight $(\mu^1, \dots, \mu^r)$. This equals λ only if every $\mu^k = \lambda^k$, so

$$T_\lambda = L(\lambda^1)_{\lambda^1} \otimes \cdots \otimes L(\lambda^r)_{\lambda^r} = \mathbb{C}\, v, \qquad v = v_1 \otimes \cdots \otimes v_r.$$

For $y_k \in U(\mathfrak{g}_k)$, the product $y_1 \cdots y_r$ acts by

$$(y_1 \cdots y_r)v = \rho_1(y_1)v_1 \otimes \cdots \otimes \rho_r(y_r)v_r.$$

Since $U(\mathfrak{g}_k)v_k = L(\lambda^k)$, the vector v generates T. The module T is finite-dimensional by Theorem 13.9 for each $\mathfrak{g}_k$, and it is the direct sum of its weight spaces. So Lemma 13.8 shows that T is simple.

Also $\mathfrak{n}_+ = \bigoplus_k \mathfrak{n}_+^{(k)}$ kills v. By (F2), T is a simple quotient of $M(\lambda)$, so $T \cong L(\lambda)$. The last sentence follows from Theorem 13.10. ∎

13.5 Example: the fundamental $\mathfrak{sl}_3$ module

For $\mathfrak{g} = \mathfrak{sl}_3$ take

$$h_1 = E_{11} - E_{22}, \quad e_1 = E_{12}, \quad f_1 = E_{21},$$
$$h_2 = E_{22} - E_{33}, \quad e_2 = E_{23}, \quad f_2 = E_{32}.$$

We record a weight by its values $(\mu(h_1), \mu(h_2))$. The brackets $[h_i, e_j] = \alpha_j(h_i)e_j$ give $\alpha_1 = (2, -1)$ and $\alpha_2 = (-1, 2)$.

The weight ω_1. Here $(m_1, m_2) = (1, 0)$. With $v = v_{\omega_1}$,

$$V(\omega_1) = M(\omega_1) \big/ \big(U(\mathfrak{g}) f_1^2 v + U(\mathfrak{g}) f_2 v\big).$$

A dominant weight $\mu = \omega_1 - a\alpha_1 - b\alpha_2$ with $a, b \in \mathbb{Z}_{\geq 0}$ satisfies

$$\mu(h_1) = 1 - 2a + b \geq 0, \qquad \mu(h_2) = a - 2b \geq 0.$$

The two inequalities give

$$4b \leq 2a \leq 1 + b, \qquad \text{so} \qquad 3b \leq 1, \quad b = 0, \quad a = 0.$$

The only dominant weight $\mu \leq \omega_1$ is ω_1. Simple reflections give

$$s_1\omega_1 = (1, 0) - \alpha_1 = (-1, 1) = \omega_1 - \alpha_1,$$
$$s_2(-1, 1) = (-1, 1) - \alpha_2 = (0, -1) = \omega_1 - \alpha_1 - \alpha_2.$$

So the set

$$S = \{(1,0), (-1,1), (0,-1)\}$$

lies in $W\omega_1$. Moreover, s_1 swaps the first two elements and fixes $(0,-1)$. Also s_2 swaps the last two and fixes $(1,0)$. So S is W-stable by (F3), and $S = W\omega_1$. By the proof of Theorem 13.9, every weight of $V(\omega_1)$ lies in this orbit. Proposition 13.7 and $\dim V(\omega_1)_{\omega_1} = 1$ give $\dim V(\omega_1) = 3$.

Compare with $\mathbb{C}^3$ and its standard basis (b_1, b_2, b_3). The diagonal matrices

$$D(h_1) = \operatorname{diag}(1,-1,0), \qquad D(h_2) = \operatorname{diag}(0,1,-1)$$

give the three weights above:

$$b_1 : (1,0), \qquad b_2 : (-1,1), \qquad b_3 : (0,-1).$$

Also $D(e_1)b_1 = D(e_2)b_1 = 0$. The lowering matrices are

$$D(f_1) = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad D(f_2) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

They give

$$D(f_1)b_1 = b_2, \qquad D(f_1)^2 b_1 = D(f_1)b_2 = 0,$$
$$D(f_2)b_1 = 0, \qquad D(f_2)D(f_1)b_1 = b_3.$$

So b_1 satisfies the defining relations. Also b_2 and b_3 lie in the submodule generated by b_1, so b_1 generates $\mathbb{C}^3$. By (F2), $v \mapsto b_1$ gives a surjection $M(\omega_1) \to \mathbb{C}^3$ killing $f_1^2 v$ and $f_2 v$, hence $N(\omega_1)$. The induced surjection $V(\omega_1) \to \mathbb{C}^3$ is an isomorphism by dimension. It sends $u, f_1 u, f_2 f_1 u$ to b_1, b_2, b_3.

Self-checks

(1) For $\mathfrak{g} = \mathfrak{sl}_2$ and $\lambda = m\omega_1$ with $m \in \mathbb{Z}_{\geq 0}$, describe $V(\lambda)$.

(2) For $\mathfrak{sl}_3$, is $2\omega_1 - 3\alpha_1$ a weight of $L(2\omega_1)$?

(3) For $\mathfrak{sl}_2$ and $\lambda = -\omega_1$, which step fails? Is $L(\lambda)$ finite-dimensional?

Answers. (1) $N(\lambda)$ is spanned by the $f^k v_\lambda$ with $k > m$, so $V(\lambda) \cong V_m$. (2) No. Its h_1-value is $2 - 6 = -4$, so s_1 sends it to $2\omega_1 + \alpha_1$, which is not $\leq 2\omega_1$. Weyl invariance excludes it. (3) The definition of $N(\lambda)$ needs $\lambda(h) \in \mathbb{Z}_{\geq 0}$, but $\lambda(h) = -1$. For the image $\bar{v}$ of v_λ in $L(\lambda)$, Lemma 13.1 gives $ef^k\bar{v} = -k^2 f^{k-1}\bar{v}$ for $k \geq 1$. If some $f^k\bar{v}$ vanished, the least such $k \geq 1$ would give $0 = -k^2 f^{k-1}\bar{v} \neq 0$. So no $f^k\bar{v}$ vanishes. Their weights differ, so they are linearly independent, and $L(\lambda)$ is infinite-dimensional.

Summary

For $\lambda \in P_+$, the finite-dimensional simple module $L(\lambda)$ is the quotient of $M(\lambda)$ by the submodule generated by $f_i^{m_i+1} v_\lambda$. Thus P_+ classifies the finite-dimensional simple modules up to isomorphism. For a sum of simple ideals, these modules are outer tensor products of simple modules of the ideals.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Section 21. Kirillov, *An Introduction to Lie Groups and Lie Algebras*, Sections 8.3 and 8.9. Carter, *Lie Algebras of Finite and Affine Type*, Section 10.3.